

Assignment No.10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
kd2_98837_sakshi> create procedure sp_rect(p_len int,p_bre int)
-> begin
-> declare v_area int;
-> declare v_perimeter int;
-> set v_area=2*(p_len+p_bre);
-> set v_perimeter=p_len*p_bre;
-> select v_area as area,v_perimeter as perimeter;
-> end;
-> $$
```

Query OK, 0 rows affected (0.01 sec)

```
kd2_98837_sakshi> delimiter ;
kd2_98837_sakshi>
kd2_98837_sakshi> call sp_rect(10,22);
```

area	perimeter
64	220

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```

kd2_98837_sakshi> delimiter $$
kd2_98837_sakshi> create procedure sp_num()
-> begin
->     declare num int;
->     set num=3;
->     insert into temp values(num,'Number');
->     insert into temp values(num*num,'Sqaure');
->     insert into temp values(num*num*num,'Cube');
-> end;
-> $$
Query OK, 0 rows affected (0.00 sec)

kd2_98837_sakshi> delimiter ;
kd2_98837_sakshi> truncate table temp;
Query OK, 0 rows affected (0.03 sec)

kd2_98837_sakshi> call sp_num();
Query OK, 1 row affected (0.01 sec)

kd2_98837_sakshi> select * from temp;
+-----+-----+
| v1    | v2      |
+-----+-----+
|      3 | Number  |
|      9 | Sqaure  |
|     27 | Cube    |
+-----+-----+
3 rows in set (0.00 sec)

kd2_98837_sakshi> |

```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```

kd2_98837_sakshi> drop procedure if exists sp_temp;
Query OK, 0 rows affected (0.01 sec)

kd2_98837_sakshi> delimiter $$
kd2_98837_sakshi> create procedure sp_temp(v_temp DOUBLE)
    -> begin
    ->     declare C DOUBLE;
    ->     declare F DOUBLE;
    ->     set C= (v_temp-32) *5/9;
    ->     set F= 9/5*v_temp + 32;
    ->     insert into temp values(C,"celcius");
    ->     insert into temp values(F,'Fahrenheit');
    -> end;
    -> $$
Query OK, 0 rows affected (0.01 sec)

kd2_98837_sakshi> delimiter ;
kd2_98837_sakshi> truncate table temp;
Query OK, 0 rows affected (0.07 sec)

kd2_98837_sakshi> call sp_temp(27);
Query OK, 1 row affected (0.02 sec)

kd2_98837_sakshi> select * from temp;
+-----+-----+
| v1          | v2          |
+-----+-----+
| -2.777777777777777 | celcius     |
|                80.6 | Fahrenheit  |
+-----+-----+
2 rows in set (0.00 sec)

```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```

kd2_98837_sakshi> delimiter $$
kd2_98837_sakshi> create procedure sp_len(p_inch int)
-> begin
->     declare v_yards int;
->     declare v_rinch int;
->     declare v_feet int;
->     set v_yards=p_inch/36;
->     set v_rinch=p_inch%36;
->     set v_feet=v_rinch/12;
->     set v_rinch=v_rinch%12;
->
->     insert into temp values(v_yards,"yards");
->     insert into temp values(v_feet,'feets');
->     insert into temp values(v_rinch,'inches');
-> end;
-> $$
Query OK, 0 rows affected (0.01 sec)

kd2_98837_sakshi> delimiter ;
kd2_98837_sakshi> truncate table temp;
Query OK, 0 rows affected (0.06 sec)

kd2_98837_sakshi> call sp_len(124);
Query OK, 1 row affected (0.02 sec)

kd2_98837_sakshi> select * from temp;
+-----+-----+
| v1    | v2    |
+-----+-----+
|      3 | yards |
|      1 | feets  |
|      4 | inches |

```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
kd2_98837_sakshi> drop procedure if exists sp_pr;
Query OK, 0 rows affected (0.00 sec)
```

```
kd2_98837_sakshi> delimiter $$
```

```
kd2_98837_sakshi> create procedure sp_pr()
```

```
    -> begin
    ->     declare v_n1 int;
    ->     declare v_n2 int;
    ->     set v_n1=10;
    ->     set v_n2=9;
    ->     if v_n1*v_n2 >=100 then
    ->         insert into temp values(v_n1*v_n2,"Product is greater than or equal to hundred");
    ->     else
    ->         insert into temp values(v_n1*v_n2,"Product is less than hundred");
    ->     end if;
    -> end;
    -> $$
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
kd2_98837_sakshi> delimiter ;
```

```
kd2_98837_sakshi> truncate table temp;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
kd2_98837_sakshi> call sp_pr();
```

```
Query OK, 1 row affected (0.00 sec)
```

```
kd2_98837_sakshi> select * from temp;
```

v1	v2
90	Product is less than hundred